DEPI PROJECT

NUVANTA

AI-Powered Enterprise Financial & Operations Management System

Evaluation & Improvements

Document	Evaluation & Improvements (Feedback, Improvements, Grading)
Project	NUVANTA
Instructor	Ahmed Mohamed Maher
Track	Full-Stack .Net Web
Team	Menna Atef, Nour Hany, Weam Emad, Nourhan Fawzy, Omar Mohamed

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1. Feedback & Evaluation

1.1 Purpose

This section records how the lecturer's assessment of NUVANTA is collected, how the team responds to it, and how the project is checked against its own targets. It is a living document: the feedback log is filled in after each review and every comment gets an action and an owner.

1.2 Feedback Channels

Channel	Who	When	Output
Lecturer review of DEPI deliverables	Instructor Ahmed Mohamed Maher	At each DEPI checkpoint	Written or verbal comments, marks
Weekly team review	Whole team; Menna Atef (Team Lead) reviews progress	Weekly, plus before each DEPI deadline	Decisions and Jira tasks
Pull request reviews	At least one reviewer per PR	Before every merge	Code comments and approvals
Usability tests	Classmates acting as users	After the end-to-end flow works (target: Week 4)	Satisfaction score and task success rate

1.3 Lecturer Feedback Log

No lecturer feedback has been recorded yet. Rows below are to be completed after each review so that every comment is traced to an action.

#	Date	Deliverable	Lecturer Feedback	Action Taken	Owner	Status
1	Week 1	Proposal and Planning (risk table)	Risk owners do not match roles: R7/R8 (AI) sit with Member 4, R3 (payments) with Member 2. Use names, not numbers.	R3 to Weam Emad; R7/R8 to Nour Hany and Menna Atef; names replace numbers.	Menna Atef	Done
2	Week 1	Project Plan – Timeline	Map Weeks 1–6 to real calendar dates and to the official DEPI deadlines.	Add date and DEPI deadline columns; confirm with the instructor.	Menna Atef	In progress
3	Week 2	Requirements Gathering	Settle company type and single vs multi-company before the SRD is frozen.	Team decision meeting; record in SRD (proposal: single company in Phase 1).	Omar Mohamed	In progress
4	Week 2–3	SRD (draft)	Make NFRs measurable: users, data volume, availability, RTO, RPO. Add IDs, priorities and acceptance criteria.	Add numeric NFR table; IDs and criteria for all Must items; start traceability matrix.	Nourhan Fawzy	Open
5	Week 3	AI features	Create seed/synthetic data early under version control. Add a moving-average baseline and report MAPE.	Seed script in repo; baseline forecast; evaluation notebook (MAPE, precision/recall).	Nour Hany	Open
6	Week 4	Payments and budgets	Write concurrency and idempotency tests for budget updates and payment confirmation (risks R3, R4).	Tests for concurrent budget updates and repeated confirmation; run in CI.	Weam Emad	Open
7	Week 4	End-to-end demo	Run the first usability test with classmates in Week 4. Add a security review checklist before final delivery.	Schedule usability session and questionnaire; write and review security checklist.	Omar Mohamed	Planned

1.4 Self-Evaluation Against Project Targets

The team checks its own work against the KPIs of the Project Planning & Management document. Results are filled in as evidence becomes available; nothing here is a measured result yet.

Area	Target (from KPIs)	Evidence	Status
Planning documents	Proposal, roles, risks and KPIs defined; timeline and requirements gathering written	Project Planning & Management, Project Plan – Timeline, Requirements Gathering	Drafted
Requirements	At least 90% of requirements covered by test cases	Final SRD and traceability matrix	Planned
End-to-end workflow	100% of the demo scenario passes	Demo checklist and automated tests	Planned
Financial accuracy	100% in reconciliation tests	Unit and reconciliation test reports	Planned
Security	100% unauthorized access blocked; 100% sensitive actions in audit log	Role-based tests; audit log review	Planned

Area	Target (from KPIs)	Evidence	Status
Performance	API under 2 s; dashboards under 3 s	API tests, browser performance tools	Planned
Code quality	Pass rate at least 95%; coverage at least 70% on financial and permission logic; 100% PRs reviewed	CI results, coverage report, GitHub history	Planned
AI features	Forecast evaluated against a baseline (MAPE); anomalies measured by precision and recall; 100% permission compliance	Evaluation notebook and permission tests	Planned
Usability	Satisfaction at least 4 of 5; task success at least 90%	Questionnaire and observed sessions	Planned

2. Suggested Improvements

The improvements below come from reviewing the NUVANTA Blueprint, the Project Planning & Management document, the Project Plan – Timeline and the Requirements Gathering document. Priority: **High** before the SRD is frozen, **Medium** during development, **Low** if time allows.

2.1 Documentation and Planning

#	Improvement	Why it helps	Priority
1	Align the risk owners with member roles. For example R7 and R8 (AI risks) are assigned to Member 4, who works on procurement and inventory, while the AI work sits with Nour Hany and Menna Atef; R3 (duplicate payments) is assigned to Member 2, while payments belong to Weam Emad.	Each risk has an owner who can actually act on it	High
2	Replace the member numbers with names in the risk table.	Avoids confusion between documents	High
3	Settle the open design decisions: company type for the workflows and single company versus multi-company.	These change requirements, database design and workflows	High
4	Map Weeks 1–6 to real calendar dates and the official DEPI deadlines.	Makes milestones checkable	High
5	Fix numeric targets in the SRD: number of users, data volume, availability, RTO and RPO.	Non-functional requirements become measurable	High
6	Build a traceability matrix linking requirement IDs to use cases and test cases.	Supports the 90% requirements-coverage KPI	Medium
7	Keep the feedback log up to date after every lecturer review.	Shows that feedback led to changes	Medium

2.2 Technical and Product

#	Improvement	Why it helps	Priority
8	Create a realistic seed and synthetic data set early (Week 2–3) and keep it under version control.	Gives the AI features enough data to evaluate and makes the demo repeatable	High
9	Define a simple baseline forecast (for example moving average) before the model, and report MAPE against it.	Shows whether the AI adds value	High
10	Write concurrency and idempotency tests for budget updates and payment confirmation as soon as the modules exist.	Covers risks R3 and R4 early	High
11	Run the first usability test with classmates in Week 4 instead of only at the end.	Leaves time to fix confusing screens	Medium
12	Add a security review checklist (authentication, authorization, input validation, error handling) before final delivery.	Supports the 100% unauthorized-access KPI	Medium
13	Add Arabic and English interface support and a supplier portal if time allows.	Closer to real users; both are ranked Should / Could	Low

2.3 Future Enhancements by Phase

These are already part of the Blueprint roadmap and stay outside the first delivery.

Phase	Enhancements
Phase 2: Business Expansion	Sales and customers, HR and payroll, product costing and production planning, wider financial reports, bank reconciliation, real payment integration if officially available
Phase 3: AI Intelligence	Demand forecasting, supplier scoring, what-if simulator, extended assistant and anomaly models
Phase 4: Enterprise Features	Multi-company and multi-branch, multi-currency, advanced workflow builder, additional integrations, data warehouse and advanced reporting

3. Final Grading Criteria

3.1 Mark Breakdown

Important: the weights below are a **proposed** split used for the team's self-assessment. They must be replaced with the official marks given by the instructor or DEPI.

Documentation 25	Implementation 40	Testing 20	Presentation 15

Category	Proposed Weight	What Is Assessed	Evidence
Documentation	25	Project proposal, planning and risks, timeline, requirements gathering, SRD with numbered requirements, diagrams, API documentation, quality and consistency	PDF documents, OpenAPI specification, GitHub README
Implementation	40	Phase 1 modules working end to end: users and roles, budgets, purchase requests and approvals, suppliers and purchase orders, goods receipt, payments (sandbox and external), ledger, inventory, audit log, dashboards; selected AI features; code quality	Working demo, GitHub repository, pull request history
Testing	20	Permission tests, financial integrity tests (budget, ledger, duplicate payments, concurrency), requirements coverage, CI pass rate, bug status at delivery	Test reports, coverage report, traceability matrix, bug tracker
Presentation	15	Clear explanation of the problem and solution, live demo of the full scenario, answers to questions, teamwork and time management	Final presentation and demo checklist
Total	**100**		

3.2 Detailed Criteria Checklist

Category	Criterion	Target	Done?
Documentation	All DEPI documents submitted on time	100%	[ ]
Documentation	Requirements have fixed IDs, priorities and acceptance criteria	All Must items	[ ]
Documentation	Architecture, ERD and OpenAPI agreed by the team	Complete	[ ]
Implementation	Demo scenario: budget, request, approval, receipt, payment, effect on budget, ledger and inventory	100%	[ ]
Implementation	Maker–Checker, RBAC and data scope enforced in the backend	100%	[ ]
Implementation	Failed payment stays unpaid; repeated confirmation not duplicated	100%	[ ]
Implementation	Forecasting, anomaly detection and AI Assistant working and explainable	3 of 3 features	[ ]
Testing	Automated test pass rate	At least 95%	[ ]
Testing	Unit coverage of financial and permission logic	At least 70%	[ ]
Testing	Requirements covered by test cases	At least 90%	[ ]
Testing	Critical bugs at final delivery	0	[ ]
Presentation	Live demo runs without failure; every member presents a part	Yes	[ ]
Presentation	Usability: satisfaction and task success	At least 4/5 and 90%	[ ]